

# The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane

$$\operatorname{Re}(s) > 7/8 - 5 \times 10^{-6}$$

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## Abstract

We examine the numerical endpoint in the compensated-probe proof of the seven-eighths zero-free half-plane in [1]. The reflected-row estimate has room to increase the total prime length, while the high-side endpoint certificate has a fixed positive margin. We extend the source moment estimates to the perturbed geometry and obtain the zero-free boundary  $174999/200000 = 0.874995$ . The accompanying Lean proof derives the moment input and the continuation conclusion from the upstream formalization.

## 1 Statement and geometry

Let  $F = \mathbb{Q}(\sqrt{-3})$ . As in [1, Equation (1.1b)], let  $\beta_*$  be the supremum of  $1/2$  and the real parts of zeros of primitive finite-order Hecke  $L$ -functions over  $F$  in the critical strip. We use the seven-eighths theorem of [1], so  $\beta_* \leq 7/8$ . We also use the arithmetic identities, detector estimates, reflected-energy lemma, additive Gram bound, and continuation criterion proved there. Section 2 checks the changed low comparison. Section 3 supplies the perturbed source moments with zero moment excess, and Section 4 checks the high comparison and analytic domains.

**Theorem 1.** *Every finite-order Hecke  $L$ -function over  $F$  and every Dirichlet  $L$ -function, including the Riemann zeta function, has no zero in*

$$\operatorname{Re} s > B := \frac{174999}{200000} = \frac{7}{8} - \frac{1}{200000}.$$

*The usual pole at  $s = 1$  for a principal character is allowed.*

Put

$$\begin{aligned} t &= \frac{3}{100000}, & \ell &= \frac{1}{6} + t, & b &= \frac{1}{8}, \\ l_x &= \frac{17}{48} - \frac{t}{2}, & l_y &= \frac{23}{48} - \frac{t}{2}, & h &= \frac{13}{16} + \frac{3t}{2}, \\ M &= l_x + l_y = \frac{5}{6} - t. \end{aligned} \tag{1}$$

These lengths satisfy  $l_y - l_x = b$ ,  $M + \ell = 1$ , and  $h = 1 - l_x + \ell$ . The general Mellin-exponent identity in [1, Part II, compensated low estimate] therefore remains

$$C(s) = \frac{l_x}{2} + s - 1 + \frac{h}{6} = s - \frac{11}{16}. \tag{2}$$

Choose finitely many distinct positive slot lengths  $\ell_j$  with  $\sum_j \ell_j = \ell$ . Their maximum can be made smaller than any prescribed positive mesh. The prime annuli at scales  $Z^{\ell_j}$  are then pairwise disjoint for all sufficiently large  $Z$ . The finite physical probe and its exact Poisson identity in [1] make sense with the lengths in (1). All its arithmetic factors are unchanged.

## 2 The low comparison

**Lemma 2.** *The normalized compensated probe of [1], formed with (1), satisfies for every  $\varepsilon > 0$*

$$|J_\eta(Z)| \ll_{\eta,\varepsilon} Z^{3/16-t/4+\varepsilon}.$$

*Proof.* Fix a rescaled subset of slots of total logarithmic length  $0 \leq d \leq \ell$ . In the notation of the compensated low estimate of [1], its row and surviving-slot lengths are

$$M' = M - 2d, \quad \ell' = \ell - d.$$

The proof of the reflected-row norm in [1, Part II, compensated low estimate] first bounds one branch by  $M'$  and the other by  $M' + (1 + 3\ell' - 2M')/4$ , up to arbitrary small losses. Its general reflected-energy lemma permits these fixed bounded lengths. Substitution gives the squared row norm

$$\sum_{0 < N \leq m \leq Z^{M'}} |B_{m,\sigma}^J(Z)|^2 \ll Z^{M' + ((d-1/6+5t)/4)_+ + \varepsilon}. \quad (3)$$

The additive Gram estimate is available throughout this range. Its scale conditions follow from

$$\begin{aligned} l_x - \ell &= \frac{3}{16} - \frac{3t}{2} > 0, & l_y - \ell &= \frac{5}{16} - \frac{3t}{2} > 0, \\ M - 2\ell &= \frac{1}{2} - 3t > 0, & l_y - \ell - \frac{11b}{6} &= \frac{1}{12} - \frac{3t}{2} > 0. \end{aligned}$$

Thus its  $P_a^2/Y'$  term is absorbed by  $P_a^{1/6}$ , exactly as in [1, Part II, additive Gram bound]. Cauchy–Schwarz, the count of rescaled tuples, and their coefficients add to the basic low exponent

$$-d + \left( \frac{d - 1/6 + 5t}{8} \right)_+.$$

This is nonpositive for  $0 \leq d \leq \ell$ . Indeed, its positive-part argument is negative until  $d > 1/6 - 5t$ , and thereafter  $d - 1/6 + 5t < 8d$ . The empty subset has exponent zero. The direct bound is consequently  $Z^{l_x/2+b/12+\varepsilon}$ .

The prime normalizer in [1, Part II, principal normalizer] is nonzero and has inverse  $Z^{o(1)}$  for every fixed positive slot length. Finally,

$$\frac{l_x}{2} + \frac{b}{12} = \frac{3}{16} - \frac{t}{4}.$$

This proves the normalized estimate. □

The signal at the proposed boundary is

$$C(B) = \frac{3}{16} - \frac{1}{200000}, \quad C(B) - \left( \frac{3}{16} - \frac{t}{4} \right) = \frac{1}{400000} > 0. \quad (4)$$

## 3 The perturbed source moments

The fourth-moment lemma with short prime factors in [1] permits  $\kappa = 3/4$ . Its zero-free hypothesis is available because  $2\beta_* - 1 \leq 3/4$ . The original detector application uses  $\kappa = 3/4 + 2\Delta$  with  $\Delta = \beta_* - 7/8 > 0$ . Here the moment estimates use  $\Delta = 0$ . The gap  $\beta_* - B$ , when positive, is used separately to choose the detector losses.

We record the estimates needed for this application. Use the normalized witness polynomials and physical slot factors of [1]. In a source fiber  $\mathcal{B}$  at scale  $U = Z^d$ , write  $r$  and  $m$  for the inverse and plain witness lengths,  $I_{u,n}$  for the inverse witness with  $n$  logarithmic derivatives, and  $P_{u,j}$  for the plain witness with  $j$  logarithmic derivatives. These include the real shifts in  $[0, 1]$  and the height shifts in  $[-H, H]$ . The physical slot factors use  $\operatorname{Re} z = 17/50$  and  $|\operatorname{Im} z| \leq H$ . For a selected set  $A$  of slots, let  $Q_A(u)$  be their physical product and put

$$\lambda_j = \ell_j/d, \quad L_A = \sum_{j \in A} \lambda_j.$$

**Lemma 3.** *Suppose  $B < \beta_* \leq 7/8$ . For every admissible choice of the perturbed detector losses, let  $\rho > 0$  be its moment loss. There is a positive mesh  $\mu$  such that slot lengths with  $\ell_j \leq \mu/200$  satisfy the following estimates. After fixing the slots and source, there is an exponent  $J \geq 0$  independent of the target character. For each target there are  $C_\eta > 0$  and a threshold such that the bounds hold for every source fiber with  $1/200 \leq d \leq 7/8$ ,  $51/100 < a \leq 7/8$ , and retained rows  $Z^{1/100} \leq Nu \leq Z^{d-\rho}$  above that threshold. With  $C_H = C_\eta(1+H)^J$ , uniformly for  $H \geq 0$ ,*

$$\sum_{u \in \mathcal{B}} |I_{u,n} Q_A(u)|^2 \leq C_H U^{1+\rho}, \quad n \leq 2, \quad r + 2L_A < 1, \quad 2r + 8L_A < 3, \quad (5)$$

$$\sum_{u \in \mathcal{B}} |P_{u,j} P_{u,k} Q_A(u)|^2 \leq C_H U^{1+\rho}, \quad j + k \leq 2, \quad 2m + \frac{9}{2}L_A \leq 1, \quad (6)$$

$$\sum_{u \in \mathcal{B}} |P_{u,j} P_{u,k}|^2 \leq C_H U^{\max\{1, 2m\} + \rho}, \quad j + k \leq 2. \quad (7)$$

The unmarked inverse raw-moment bounds also hold with the count parameters chosen from the fixed source. Together these are the source moment bounds with zero moment excess required by the detector.

*Proof.* Apply the certified energy theorem of [1] with  $\kappa = 3/4$  and stage error  $\rho/4$ . Its positive-width and zero-slot conclusions give a positive fine mesh  $\mu$ . The condition  $\ell_j \leq \mu/200$  ensures  $\lambda_j \leq \mu$  for every  $d \geq 1/200$ . This does not restrict the total slot length, since the number of distinct positive slots can be increased.

Fix the source's finite excluded prime set  $S$ , and let  $M_S$  be its product ideal. Use the fixed ideal  $M_S \cap (72)$  in the energy source. For each target there are only finitely many source labels. The norm of the character modulus at every such label is at most  $U^{\rho/4}$  once  $Z$  is large enough. The exceptional sixth-power-free rows are supported on a fixed finite set of primes and have bounded norm. They are therefore absent from the retained rows for large  $Z$ .

The source identities bound the sum of the squared plain witness products by retained energy with the physical prime coefficients. The marked initial state has width  $1 + \rho/4$ . Its capacity condition follows from  $2m + 6\kappa L_A \leq 1$ , and the positive-width energy bound gives exponent  $1 + \rho/2 \leq 1 + \rho$ . For the unmarked pair, use initial width  $\max\{1, 2m\} + \rho/4$  and the zero-slot energy bound. This gives (6) and (7).

For the inverse witness, the fixed-source inverse estimates of [1] apply under the two strict capacity inequalities in (5). Distinct slot lengths give disjoint prime supports. Their fixed conductor deletions have modulus-one denominators on these supports for large  $Z$ . The inverse raw-moment bound and the marked inverse bound consequently apply to the perturbed slots.

The profile estimates bound all needed seminorms by a fixed power of  $1 + H$ . Taking a common exponent and summing the finitely many label constants gives  $J$  and  $C_\eta$ . The exponent is chosen before the target. Thus the estimates hold at detector height  $H = Z^{2\tau}$ ,

and the usual late choice of  $\tau > 0$  absorbs the polynomial height factor into the prescribed detector loss.  $\square$

## 4 The high comparison

We first record the extension of the Euler regions used by the contour arguments. The local factors below are those of [1, Equations (5.13a)–(5.13c)].

**Lemma 4.** *The full compensated correction is holomorphic, with the bounds needed for principal-row contour moves, on the second Euler region of [1] with its condition  $\operatorname{Re} s \geq 7/8$  replaced by  $\operatorname{Re} s \geq B$ . The principal local quotient there satisfies*

$$G_p/H_p = -1 + O((Np)^{-B}).$$

*The small-row contour can use the first Euler region with  $\epsilon_0 = 37/100$ .*

*Proof.* The exact local defect formula in [1, local Euler lemma] bounds  $H_p - 1$  in the original second region by  $O(Q^{-363/200})$  for  $p \nmid u$  and  $O(Q^{-33/40})$  for  $p \mid u$ , where  $Q = Np$ . Every local exponent involving  $\operatorname{Re} s$  has coefficient of absolute value at most six. Moving its lower boundary left by  $1/200000$  therefore gives the respective bounds

$$O(Q^{-363/200+3/100000}) \quad \text{and} \quad O(Q^{-33/40+3/100000}).$$

The first exponent is less than  $-1$ . The unramified product converges normally, while the ramified product has the same  $O_\varepsilon((Nu)^\varepsilon)$  bound. For  $u = 1$  its tail still tends to zero faster than  $P_0^{-4/5}$ . Enlarging the fixed excluded set therefore gives the same uniform nonvanishing of the principal product.

The four errors for the principal selected factor have exponents  $-\operatorname{Re} s$ ,  $-6 \operatorname{Re} z$ ,  $4 - 5 \operatorname{Re} s - 6 \operatorname{Re} z$ , and  $1 - \operatorname{Re} w - 6 \operatorname{Re} z$ . On the extended region they are all at most  $-B$ . The formula for  $G_p$  has no division by  $P_p$  or  $H_p$ , so the full finite selected-tuple correction remains holomorphic at their zeros. The bounds above and finite slot count give the all-height majorant used in the principal contour proof of [1].

For small rows,  $\beta_* > B$  implies  $\beta_* + e + 1/2 > 1 + 37/100$ . The first Euler region of [1] allows any positive  $\epsilon_0$ , so its unchanged local estimates apply to the line  $(\beta_* + e, 1/2, 17/50)$ . The central bin contours already lie in its region with  $\epsilon_0 = 10e$ . These are the only analytic domain uses of the original  $7/8$  lower bound in the shared contour lemmas. Repeating their proofs with the extended region gives the same principal, bin, and outer-row contour estimates.  $\square$

**Lemma 5.** *If  $B < \beta_* \leq 7/8$ , then for some  $\sigma > 0$  independent of the target character,*

$$|J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta_*)-\sigma},$$

*where  $f_\eta$  is the principal Mellin signal from [1] with (2).*

*Proof.* Lemma 3 supplies the fourth-moment estimate at  $\kappa = 3/4$ , because its zero-free input is  $\beta_* \leq (1 + \kappa)/2 = 7/8$ . Every detected bin has  $\delta = 2a - 1 \leq 2\beta_* - 1 \leq 3/4$ . Its plain prime capacity is thus the baseline capacity in the proof of [1, Part II, capacity comparison], with no positive  $\Delta/4$  surcharge. The row exponent is bounded by  $R_*(\delta, x)$  plus an arbitrarily small loss, with  $0 \leq x \leq 1/2$  and  $q = x\delta$ . The same selected-prime argument applies since  $\ell/h > 8/39 > 7/37$  and  $5\ell - h > 1/48$ . A fixed sufficiently fine even slot mesh meets all rounding conditions.

At row length  $d = h$ , the general high exponent relative to  $C(B)$  is

$$E_B(h) = a - B + h \left( \frac{17}{50} - \frac{1}{6} \right) - a l_y - \frac{\ell}{2} + q \ell + h \left( R + \frac{\delta}{2} - \frac{17}{50} \right). \quad (8)$$

The endpoint certificate of [1, Part II, compensated endpoint certificate], evaluated at its original geometry, bounds this expression with  $B = 7/8$  and  $R = R_*$  by  $-49/440640$  for the larger range  $\delta \leq 5/6$ . Holding  $a, \delta, q, R$  fixed while changing (1) gives

$$E_B(h) - E_{7/8}(13/16) = \frac{1}{200000} + t \left( -\frac{1}{2} + \delta + q + \frac{3R}{2} \right). \quad (9)$$

The source bounds  $R_* \leq 17/12$ . Since  $\delta \leq 3/4$  and  $q \leq \delta/2$ , the right side of (9) is at most

$$\frac{1}{200000} + \frac{11t}{4} = \frac{7}{80000}.$$

Thus the central endpoint retains the explicit margin

$$-E_B(h) \geq \frac{49}{440640} - \frac{7}{80000} > 0. \quad (10)$$

The frequency slope is positive, so this endpoint bounds all moderate rows with  $d \leq h$ . Extending to  $h + \zeta$  costs at most  $2\zeta$ , and  $\zeta$  can be chosen below the margin in (10).

The floor and intermediate-row margins in [1] are respectively  $7/1200$  and  $49/14400$  at the original geometry. The floor shift costs  $t(121/40 + 3e + \text{mesh})$ , including the changed row threshold. The parameter budget absorbs this cost. The small-row margin is  $63/800$ . Its change is at most  $51t/100$ , since  $h(17/50 - 1/6) - l_y/2$  has derivative  $51/100$  with respect to  $t$ . For large rows, the absolute contour can still be taken sufficiently far right after choosing  $\zeta$ . Lemma 4 supplies the principal contour, whose margins  $l_y/20$  and  $h/600$  remain positive. Its local approximation supplies a positive power saving in each fixed slot. The prime normalizer costs only  $Z^{o(1)}$ .

All detector, mesh, real-contour, and normalizer losses can be chosen below these fixed positive margins before the target character is fixed. The late height choice in [1] then gives one  $\sigma > 0$  for every target, as claimed.  $\square$

*Proof of Theorem 1.* Suppose  $\beta_* > B$ . Lemma 4 supplies a principal factor holomorphic and bounded away from zero on  $\text{Re } s > B$ . Lemma 2 and (4) give the direct bound required by the continuation criterion of [1, Section 2]. Its parameter  $\omega$  can be any sufficiently small positive number below  $\beta_* - B$ . Lemma 5 gives the other required bound with a target-independent saving. The criterion contradicts  $\beta_* > B$ . Hence  $\beta_* \leq B$ . The transfer in [1] from primitive Hecke characters to imprimitive Hecke and Dirichlet characters proves the stated assertions.  $\square$

## Verification scope

The rational inequalities in (4) and (10) can be checked independently by the accompanying `verify_bounds.py`. The accompanying Lean files verify the geometry, reflected-exponent calculation, high-side endpoint, normalized compensated low estimate, local Euler bounds, and principal and global Euler product extensions against the upstream OAI formalization. The finite source multiplier bounds and the principal source's  $w$  and  $z$  contour shifts, ordered double shift, uniform contour tails, and  $x$  contour transport also have Lean proofs at the new boundary. The selected high-row local factors, weighted prime-slot sum, physical-row dyad, Mellin integral, and changed small- and large-row tail bounds, central cube truncation

error, canonical ray-cube choice, and normalized floor-row collection are verified there as well. The principal slot-ratio and normalized residue estimates are proved on the new boundary. The physical principal remainder at the changed lengths is bounded on a fixed contour just right of  $7/8$ , leaving a power saving of  $1/3000$  relative to  $\beta_*$  in the proposed interval. The perturbed source moment transport is proved in `OAIHighMomentTransport.lean`. It supplies the chosen fourth-moment input used by the central-bin estimate and detector assembly. `OAIHighUnconditional.lean` derives the prior  $\beta_* \leq 7/8$  bound from the upstream unconditional module and concludes the stronger Hecke, Dirichlet, and zeta nonvanishing statements without a moment hypothesis. The final theorem’s axiom audit reports only the standard Lean axioms `propext`, `Classical.choice`, and `Quot.sound`. The proof uses Lean 4.34.1 and upstream revision `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Compiling this note checks the TeX. The separate Lean build checks the proof. The stronger boundary in Theorem 1 is an extension of [1]’s stated theorem.

## References

- [1] OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane*  $\operatorname{Re}(s) > 7/8$ , 30 September 2026. Source consulted 7 October 2026. [Source TeX at the verified revision](#).